Raft Docker Lab

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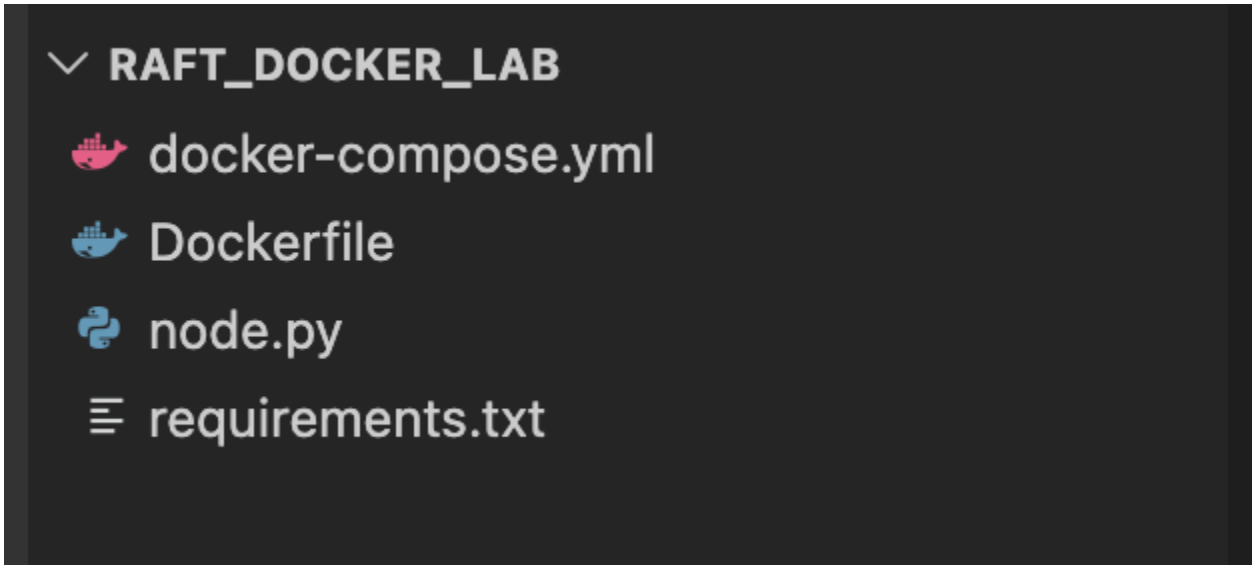
Sem 6 ‘E’

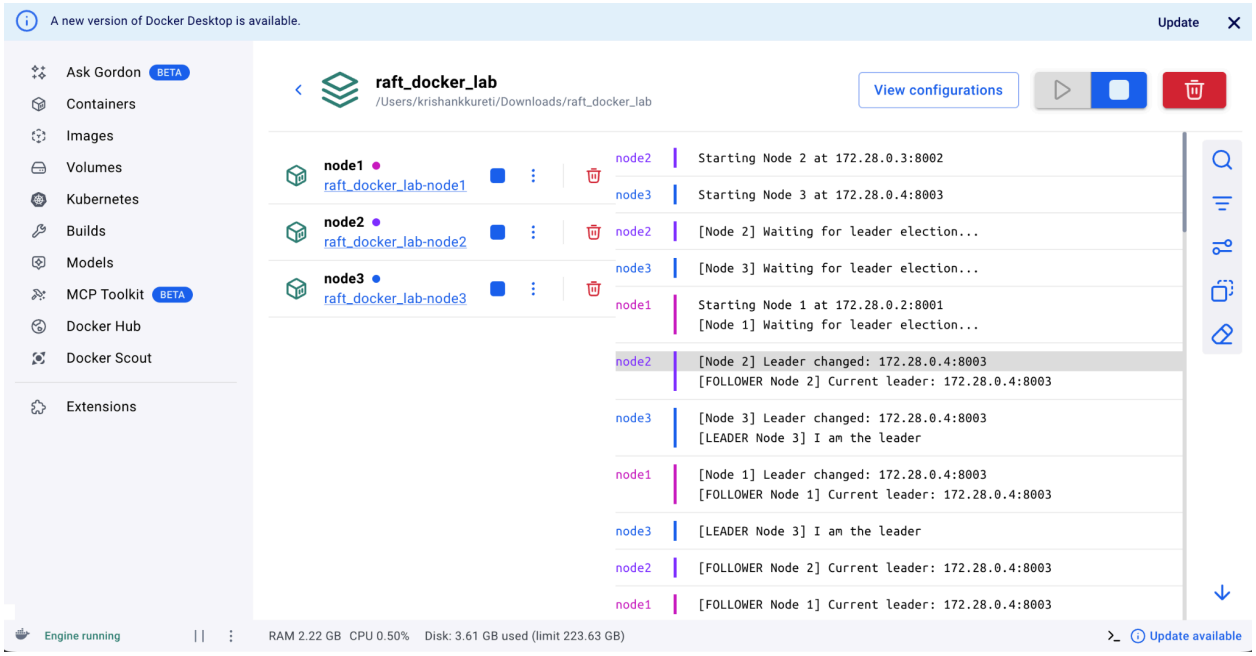
Ss1



Ss2a

```
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker compose up --build
 [+] Building 36.3s (21/21) FINISHED
  => [internal] load local bake definitions                                                                    0.0s
  => => reading from stdin 1.50kB                                                                              0.0s
  => [node1 internal] load build definition from Dockerfile                                                    0.0s
  => => transferring dockerfile: 489B                                                                          0.0s
  => [node1 internal] load metadata for docker.io/library/python:3.7.16-bullseye                               5.1s
  => [auth] library/python:pull token for registry-1.docker.io                                                 0.0s
  => [node3 internal] load .dockerignore                                                                       0.0s
  => => transferring context: 2B                                                                               0.0s
  => [node3 1/9] FROM docker.io/library/python:3.7.16-bullseye@sha256:d98fd1fa8fd1756bfe883b5004a60cfc428a1ce6e575652b771b66c8  19.1s
  => => resolve docker.io/library/python:3.7.16-bullseye@sha256:d98fd1fa8fd1756bfe883b5004a60cfc428a1ce6e575652b771b66c8db22247  0.0s
  => => sha256:37e42497053dc37f73831a394f66536bf7a82f72808951ad6945716ac4b95c00 2.91MB / 2.91MB                0.8s
  => => sha256:0c2832b2823a30a0e4aad22add0a8ca59cb78915b8077b3188579342c6590a86 14.81MB / 14.81MB              3.1s
  => => sha256:ae66119f5d3a05ac439f95c40d5041bb518640b1b348e22498ee478d87ec7348 245B / 245B                    0.9s
  => => sha256:4d727a249b449ca216ff6e0b6f53a712285395ca26c1ad7ed98aebe0548bf250 6.40MB / 6.40MB                1.3s
```

Ss2b



Ss3a

```
node3   [LEADER Node 3] I am the leader
node2   [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node1   [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

Ss3b

```
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
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[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

Ss3c

```
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
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[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

Ss3d

```
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

~/Downloads/raft_docker_lab — docker logs -f node3

Ss4

```
Last login: Fri Mar 27 14:51:59 on ttys005
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker ps
CONTAINER ID   IMAGE                    COMMAND              CREATED         STAT
US         PORTS     NAMES
8213f0ac427d   raft_docker_lab-node1   "python node.py 1"   8 minutes ago   Up 8
 minutes             node1
9d4e2503d63f   raft_docker_lab-node2   "python node.py 2"   8 minutes ago   Up 8
 minutes             node2
f18165b0d9bd   raft_docker_lab-node3   "python node.py 3"   8 minutes ago   Up 8
 minutes             node3
krishankkureti@Krishanks-MacBook-Air raft_docker_lab %
```

Ss5

```
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker network inspect raft_docker_lab_raftnet
[
    {
        "Name": "raft_docker_lab_raftnet",
        "Id": "470ced4152bfbb1d0421e618f022b676d6d2f385af189264f5573985be2a1c7d",
        "Created": "2026-03-27T09:17:45.034532462Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.28.0.0/16",
                    "Gateway": "172.28.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {
            "com.docker.compose.config-hash": "7b955128463677559d4edba13e9fe8f69dd803b2dfc190dd0f2333946fc6a5d0",
            "com.docker.compose.network": "raftnet",
            "com.docker.compose.project": "raft_docker_lab",
            "com.docker.compose.version": "5.0.2"
        },
        "Containers": {
            "8213f0ac427d78377d2cf5c7e0621799d6fedd7116a4bc1a22b663d8bbae69b7": {
                "Name": "node1",
                "EndpointID": "de90afa3dc98152269e71e413264645cd077531d7cd3292f7e8f3cba2a946da0",
                "MacAddress": "6a:d9:77:f6:fb:75",
                "IPv4Address": "172.28.0.2/16",
                "IPv6Address": ""
            },
            "9d4e2503d63fd67276364eb7c950214eca41636c0de4ec41451353e64eb73d77": {
                "Name": "node2",
                "EndpointID": "b142d5a41a231d0857c6498f757bd65105d27bfc2d81a00d767ccbb41d297fa8",
                "MacAddress": "0a:f2:33:c2:4e:20",
                "IPv4Address": "172.28.0.3/16",
                "IPv6Address": ""
            },
            "f18165b0d9bd84c293eed4bbd180dd990560b5d7ec28144c0909b7edbfae6dc9": {
                "Name": "node3",
                "EndpointID": "96c5fd38b7423e14edb5fc98d8cb805bf31aaf15e9ed45b5846dc09fd89c7fb2",
                "MacAddress": "e2:d3:db:33:9b:2e",
                "IPv4Address": "172.28.0.4/16",
                "IPv6Address": ""
            }
```

Ss6a

```
node1   [Node 1] Leader changed: None
        [Node 1] Waiting for leader election...

node2   [Node 2] Leader changed: None
        [Node 2] Waiting for leader election...

node1   [Node 1] Leader changed: 172.28.0.3:8002
        [FOLLOWER Node 1] Current leader: 172.28.0.3:8002

node2   [Node 2] Leader changed: 172.28.0.3:8002
        [LEADER Node 2] I am the leader
        [LEADER Node 2] I am the leader

node1   [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
        [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

Ss6b

```
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
```

Ss6c

```
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

Ss7a

```
node3   [Node 3] Leader changed: 172.28.0.3:8002
        [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node2   [LEADER Node 2] I am the leader
node1   [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node3   [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node2   [LEADER Node 2] I am the leader
node1   [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node3   [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

Ss7b

```
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.4:8003
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

Ss8

```
Last login: Fri Mar 27 14:56:06 on ttys006
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker ps
CONTAINER ID   IMAGE                  COMMAND              CREATED          STATUS          PORTS     NAMES
c85b65d4d048   raft_docker_lab-node3  "python node.py 3"   19 seconds ago   Up 19 seconds             node3
bafdbb5670f0   raft_docker_lab-node2  "python node.py 2"   19 seconds ago   Up 19 seconds             node2
d1b8fb70305f   raft_docker_lab-node5  "python node.py 5"   19 seconds ago   Up 19 seconds             node5
e598d6a93c35   raft_docker_lab-node4  "python node.py 4"   19 seconds ago   Up 19 seconds             node4
4c19e23420ca   raft_docker_lab-node1  "python node.py 1"   19 seconds ago   Up 19 seconds             node1
krishankkureti@Krishanks-MacBook-Air raft_docker_lab %
```

Ss9

```
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

Ss10

```
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker logs -f node2
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

Ss11

```
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.4:8003
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

Ss12

```
[Node 4] Waiting for leader election...
[Node 4] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
[FOLLOWER Node 4] Current leader: 172.28.0.4:8003
```

Ss13

```
Starting Node 5 at 172.28.0.6:8005
[Node 5] Waiting for leader election...
[Node 5] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 5] Current leader: 172.28.0.4:8003
[FOLLOWER Node 5] Current leader: 172.28.0.4:8003
[FOLLOWER Node 5] Current leader: 172.28.0.4:8003
[FOLLOWER Node 5] Current leader: 172.28.0.4:8003
[FOLLOWER Node 5] Current leader: 172.28.0.4:8003
[FOLLOWER Node 5] Current leader: 172.28.0.4:8003
```

Ss14

```
Last login: Fri Mar 27 15:17:35 on ttys008
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED         STATUS         PORTS     NAMES
c85b65d4d048   raft_docker_lab-node3   "python node.py 3"   7 minutes ago   Up 7 minutes             node3
bafdbb5670f0   raft_docker_lab-node2   "python node.py 2"   7 minutes ago   Up 7 minutes             node2
4c19e23420ca   raft_docker_lab-node1   "python node.py 1"   7 minutes ago   Up 7 minutes             node1
krishankkureti@Krishanks-MacBook-Air raft_docker_lab %
```

Ss15

```
krishankkureti@Krishanks-MacBook-Air raft_docker_lab % docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED          STATUS          PORTS     NAMES
c85b65d4d048   raft_docker_lab-node3   "python node.py 3"   10 minutes ago   Up 10 minutes             node3
bafdbb5670f0   raft_docker_lab-node2   "python node.py 2"   10 minutes ago   Up 10 minutes             node2
```

Ss16

```
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
```

Ss17

```
[LEADER Node 3] I am the leader
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Waiting for leader election...
[Node 3] Waiting for leader election...
[Node 3] Waiting for leader election...
```

Why did leader election not occur when only 2 nodes were active in a 5-node cluster?

Because Raft doesn’t allow a leader unless a majority agrees. With only 2 nodes alive in a 5-node cluster we don’t have enough nodes for a majority.